

Customer Satisfaction Project

Student Project Handout: Regression, Confidence Intervals, Normality, Data Reshaping, Pivot Tables, and Visualization

Project Overview

In this project, you will act as a customer experience analyst examining which service and customer characteristics are associated with satisfaction outcomes. Using the provided Excel workbook, you will reshape data, build and interpret linear regression models, evaluate normality, compute and interpret confidence intervals, produce a descriptive Table 1 with pivot tables, and create compelling visualizations.

Learning Goals

- Use Excel and statistical reasoning to explore a customer satisfaction dataset.
- Fit and interpret at least one linear regression model with a continuous outcome.
- Interpret statistical significance, confidence intervals, and model results in plain language.
- Reshape data from wide to long format using Excel Power Query.
- Create a descriptive Table 1 using pivot tables.
- Develop audience-friendly customer experience visualizations.

Dataset

The workbook contains 2,000 rows. The primary worksheet is **Customer_Satisfaction_Wide**. The repeated quarterly variables **Satisfaction_Q1**, **Satisfaction_Q2**, **Satisfaction_Q3**, and **Satisfaction_Q4** are included specifically for the wide-to-long transformation task.

Project Tasks

- 1. Initial data review.** Review the data dictionary and identify one outcome variable and several possible predictors.
- 2. Wide-to-long transformation.** Use Excel Get Data / Power Query to pivot Satisfaction_Q1 through Satisfaction_Q4 into a long format dataset with one variable identifying quarter and one variable containing satisfaction scores.
- 3. Table 1 using pivot tables.** Create a descriptive summary table for selected variables. Include counts for categorical variables and descriptive statistics for quantitative variables. At a minimum, report counts, means, medians, standard deviations, and IQRs for several key variables.
- 4. Linear regression.** Fit at least one linear regression model with **Customer_Satisfaction_Score** as the response. You may also explore **Recommendation_Score** as an alternate outcome. Include at least three predictors.
- 5. Interpret significance.** Explain the meaning of the coefficient estimates, p-values, and overall findings. Identify which predictors appear statistically significant and explain what that means in context.
- 6. Explore normality.** Evaluate the distribution of at least one quantitative variable and the regression residuals. Use plots and/or descriptive evidence to discuss whether normality appears reasonable.

7. Confidence intervals. Compute and interpret at least one confidence interval for a mean and at least one confidence interval connected to a regression estimate.

8. Visualizations. Create at least three polished visuals that communicate meaningful patterns in the data.

Possible Research Questions

- Are shorter wait times associated with higher customer satisfaction?
- How is sentiment score related to satisfaction outcomes?
- Do support channels differ in average satisfaction score?
- Is the number of contacts needed associated with lower customer satisfaction?
- Do issue types differ in average satisfaction or recommendation score?

Suggested Deliverables

- A cleaned Excel workbook that includes the wide-to-long transformation output.
- A Table 1 worksheet or screenshot showing descriptive statistics from pivot tables.
- A short written report or slide deck summarizing your regression results, confidence intervals, and interpretation.
- At least three polished visualizations.
- A brief conclusion explaining the most important findings and any limitations.

Minimum Variables to Consider

Variable	Role in Project
Customer_Satisfaction_Score	Suggested regression outcome
Recommendation_Score	Alternative regression outcome
Wait_Time_Min / Resolution_Time_Min	Quantitative predictors
Number_of_Contacts	Quantitative predictor
Purchase_Value_USD / Response_Time_Hours	Quantitative predictors
Agent_Experience_Years / Sentiment_Score	Quantitative predictors
Support_Channel / Customer_Type / Issue_Type	Categorical predictors or grouping variables
Satisfaction_Q1 to Satisfaction_Q4	Wide-to-long reshaping task

Submission Checklist

- Long-format data created correctly from quarterly satisfaction columns
- Table 1 completed with pivot tables
- Regression model shown and interpreted correctly

- Confidence intervals reported and interpreted in words
- Normality discussion included
- At least three polished visualizations included
- Final conclusions written for a customer experience audience